

**(Fluidized Beds)**

Pre-Lab Plan

Unit Operations Lab 1

01/20/2026

Group 1, Tuesday, AM section

*Team Member Names:*

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## 1. Experimental Objective (5 pts)

The objective of this experiment is to study gas–solid fluidization by measuring the pressure drop across a bed of particles as a function of superficial gas velocity. Two different bed materials and two static bed heights are tested to compare how particle properties and bed depth affect fluidization behavior. Average particle size is determined using sieve analysis, and the bed void fraction is calculated to support interpretation of the flow data. From the pressure-drop curves, the minimum fluidization velocity is identified and compared with values predicted from correlations. Overall, the experiment examines the transition from packed-bed to fluidized-bed conditions and relates measured operating data to key fluidization parameters.

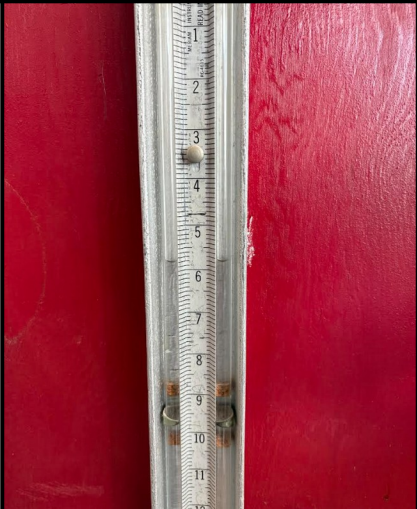
## 2. Experimental

### 2.1 Apparatus (5 pts)

In this experiment, there are only a few key pieces of equipment. Among these is the packed bed itself, the pressure needle valve, pressure gauges, flowmeters, and directional valves. The complete list of equipment, along with images and brief descriptions of their functions, is provided in Table 1.

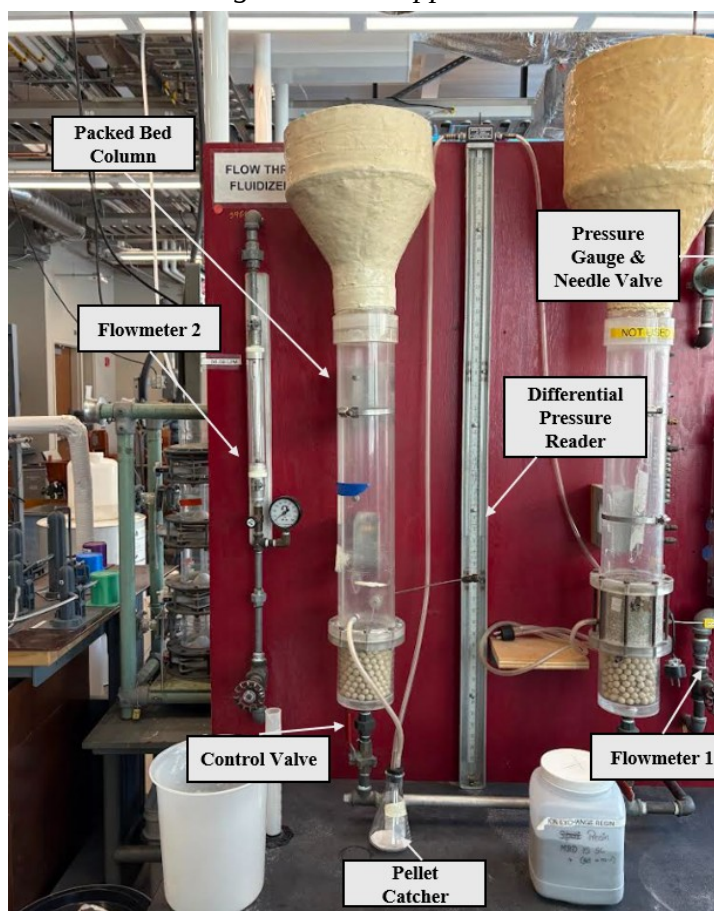
*Table 1 – Laboratory Equipment*

| Equipment             | Number | Picture                                                                              | Description                                                           |
|-----------------------|--------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Pressure Needle Valve | 1      |   | Nozzle that enables air from utilities to flow into the system.       |
| Pressure Gauge        | 2      |  | Equipment used to control the pressure of air flowing into the system |

|                                                   |                 |                                                                                      |                                                                                                                                                                                |
|---------------------------------------------------|-----------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Flowmeter 1</b></p>                         | <p><b>3</b></p> |    | <p><b>Equipment that controls and measures the flow of air entering the packed bed column at high flowrates.</b></p>                                                           |
| <p><b>Differential Pressure Reader</b></p>        | <p><b>4</b></p> |   | <p><b>Equipment used to measure pressure drop inside the packed bed column.</b></p>                                                                                            |
| <p><b>Shut-Off Valve &amp; Pellet Catcher</b></p> | <p><b>5</b></p> |  | <p><b>The valve is used to send or stop air from entering the column. The flask collects catalyst pellets that may get caught inside the differential pressure reader.</b></p> |

|                                 |                 |                                                                                     |                                                                                                                                |
|---------------------------------|-----------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Flowmeter 2</b></p>       | <p><b>6</b></p> |   | <p><b>Another piece of equipment used to control and measure the flow of air entering the column at smaller flowrates.</b></p> |
| <p><b>Packed Bed Column</b></p> | <p><b>7</b></p> |  | <p><b>Column used to test fluidization of different pellets under varying flowrates of air.</b></p>                            |

Figure 1: Full Apparatus

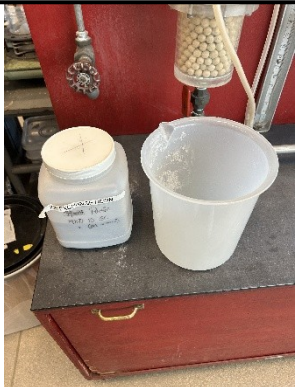
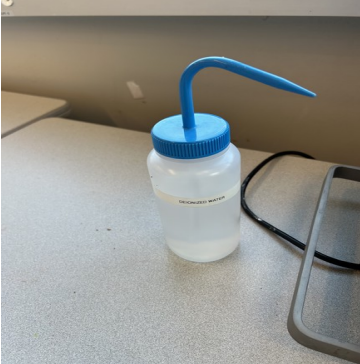
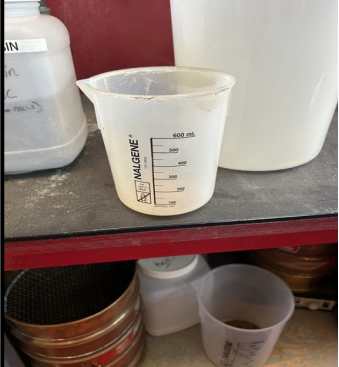


## 2.2 Materials and Supplies (5 pts)

The complete material and supply list required for the fluidized bed experiment is provided in Table 2. This table includes all materials and supplies used in the experiment, along with brief details for each item.

Table 2: Material and Supply List

| Equipment | Number | Picture | Description |
|-----------|--------|---------|-------------|
|-----------|--------|---------|-------------|

|                                 |          |                                                                                      |                                                                                                                                              |
|---------------------------------|----------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ion-exchange resin beads</b> | <b>1</b> |    | <b>Modules housing bundles of hollow fibers in which selective permeation occurs to produce oxygen-enriched and oxygen-depleted streams.</b> |
| <b>Metal sieves</b>             | <b>2</b> |    | <b>Stack of metal sieves used to separate resin or sand by particle size</b>                                                                 |
| <b>DI water</b>                 | <b>3</b> |   | <b>Bottle filled with deionized water used to add water to a flask during void fraction measurements.</b>                                    |
| <b>Plastic Graduated Beaker</b> | <b>4</b> |  | <b>Graduated containers used to measure approximate volumes.</b>                                                                             |

|                           |          |                                                                                   |                                                                                                          |
|---------------------------|----------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Scale</b>              | <b>5</b> |  | <b>Used to measure the mass of the sieve stack with and without resin and void fraction measurements</b> |
| <b>Graduated cylinder</b> | <b>6</b> |  | <b>Used to add water to resin for void fraction measurements</b>                                         |

### 2.3 Experimental Plan (10 pts)

#### Procedure

- 1) Turn on the air supply to the fluidized bed apparatus and ensure all valves are initially closed
- 2) Perform a sieve analysis for each bed material (fine silica sand mixture and ion-exchange resin beads) to determine the particle size distribution and calculate the average particle diameter.

- 3) Charge the fluidized-bed column with the selected bed material to a static packed-bed height of 6 inches. Ensure the bed is level and uniformly distributed above the distributor plate.
- 4) Begin airflow at a very low rate and gradually increase the gas flow using the control valves. At each flowrate, allow the system to stabilize and record the superficial gas velocity and the pressure drop across the bed. Continue increasing the airflow until the bed reaches the fully fluidized regime which is when the pressure drop becomes approximately constant with increasing flowrate.
- 5) After reaching the maximum airflow, decrease the gas flow incrementally back to zero. At each step, record the gas flowrate and pressure drop to capture hysteresis behavior.
- 6) Repeat the increasing- and decreasing-flow procedure for a packed-bed height of 10 inches using the same bed material.
- 7) Remove the first bed material and repeat the entire procedure for the second bed material (ion-exchange resin beads), conducting tests at both 6-inch and 10-inch bed heights.
- 8) After completing all runs, shut off the air supply, slowly depressurize the system, and leave all valves in safe positions.

## **Variables**

### Independent Variables

- Superficial gas velocity (or gas flowrate)
- Bed material type (sand or ion-exchange resin)
- Packed-bed height (6 in, 10 in)
- Flow direction (increasing vs decreasing)

### Controlled Variables

- Column geometry and distributor plate
- Fluidizing gas (air)
- Ambient temperature and pressure

### Dependent Variables

- Pressure drops across the bed
- Pressure drops per unit bed height
- Minimum fluidization velocity
- Bed hysteresis behavior

### Data and Model Correlation

- Particle size is obtained from sieve analysis and used to compute the average particle diameter.
- The pressure drop in the packed-bed regime is modeled using the Ergun equation:

$$\frac{\Delta P}{L} = \frac{150 \mu \bar{V}_0 (1-\varepsilon)^2}{\Phi_s^2 D_p^2 \varepsilon^3} + \frac{1.75 \rho \bar{V}_0^2 (1-\varepsilon)}{\Phi D_p \varepsilon^3}$$

- Minimum fluidization velocity is determined experimentally as the superficial velocity at which the pressure drop becomes constant with increasing flowrate.
- Hysteresis is evaluated by comparing pressure-drop curves obtained during increasing and decreasing airflow conditions.

### Plots to Generate

- Pressure drop per unit bed height versus superficial gas velocity for increasing and decreasing flow
- Identification of minimum fluidization velocity for each bed material and bed height
- Comparison of experimental packed-bed data with Ergun equation predictions
- Particle size distribution from sieve analysis
- Comparison of fluidization behavior between sand and ion-exchange resin beds

### Project Safety Evaluation (10 pts)

|                               | Yes/No |                                                                                                                                      |
|-------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------|
| Potential electrical hazards? | No     | If yes, identify conditions:                                                                                                         |
| Potential mechanical hazards? | No     | If yes, identify conditions:                                                                                                         |
| Potential pressure hazards?   | Yes    | If yes, identify conditions: Compressed air up to 120 psig from the wall (regulated to ~40 psig) could cause hose whip or sudden air |

|                                     |                           |                                                                                                                                                                                                                                                 |
|-------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     |                           | release if the regulator, tubing, or fittings fail or are opened suddenly. Depressurizing (reducing air flow rate) should happen gradually.                                                                                                     |
| Potential temperature hazards?      | No                        | If yes, identify conditions:                                                                                                                                                                                                                    |
| Hazardous/toxic chemicals involved? | No                        | If yes, hazards involved:                                                                                                                                                                                                                       |
| Gloves required?                    | No                        | If yes, specify type:                                                                                                                                                                                                                           |
| MSDS reviewed by all team members?  | Yes                       | List MSDS sheets reviewed: <a href="#">Sand</a><br><a href="#">Ion-exchange resin</a>                                                                                                                                                           |
| <i>Process Parameters</i>           | <i>Max Expected</i>       | <i>List location and possible hazards and precautions</i>                                                                                                                                                                                       |
| Temperature (°C)                    | ~25 °C (room temperature) | The whole system is near room-temperature, and there are no temperature controllers to adjust temperatures.                                                                                                                                     |
| Pressure (psia)                     | 120 psig (wall)           | Air comes in at 120 psig from the wall and is reduced to about 40 psig through a pressure regulator before entering the column.                                                                                                                 |
| <i>Safety Controls</i>              | <i>yes/no</i>             | <i>If yes, Provide description</i>                                                                                                                                                                                                              |
| Slipping                            | yes                       | The ion-exchange resin beads can be slippery if spilled on the floor, so spills should be cleaned up (vacuumed) quickly to avoid slips.                                                                                                         |
| Pressure management                 | yes                       | The wall supplies compressed air at about 120 psig, which is reduced to about 40 psig using a pressure regulator before entering the system. Flow is adjusted gradually using the flow controls to avoid sudden pressure changes in the column. |
| Irritant management                 | yes                       | Sand and dust can irritate the eyes and nose when handled or poured. Goggles should be worn, and a mask is recommended if noticeable dust is present.                                                                                           |

I have read relevant background material for the Unit Operations Laboratory entitled: “*Fluidized beds*” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Joshua Olsuji 01/20/2026  
signature of team member date

Omar Zaruk 01/20/2026  
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